# Deep Learning in Drug Target Interaction Prediction: Current and Future Perspective

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REVIEW ARTICLE

# Deep Learning in Drug Target Interaction Prediction: Current and Future Perspective

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**Abstract:** Drug-target Interactions (DTIs) prediction plays a central role in drug discovery. Computational methods in DTIs prediction have gained more attention because carrying out *in vitro* and *in vivo* experiments on a large scale is costly and time-consuming. Machine learning methods, especially deep learning, are widely applied to DTIs prediction. In this study, the main goal is to provide a comprehensive overview of deep learning-based DTIs prediction approaches. Here, we investigate the existing approaches from multiple perspectives. We explore these approaches to find out which deep network architectures are utilized to extract features from drug compound and protein sequences. Also, the advantages and limitations of each architecture are analyzed and compared. Moreover, we explore the process of how to combine descriptors for drug and protein features. Likewise, a list of datasets that are commonly used in DTIs prediction is investigated. Finally, current challenges are discussed and a short future outlook of deep learning in DTI prediction is given.

**Keywords:** Drug-target interaction prediction, Deep learning, Machine learning, Drug discovery, DTIs prediction approaches.

## 1. INTRODUCTION

Drug-target interactions (DTIs) has a vital role in the drug discovery process. DTIs identify the interaction sites between drug compounds and protein targets and characterize the properties of the interactions sites. The main goal of DTI is to identify new ligands against specific protein targets. Many research areas have gotten benefits from identifying DTIs, including drug repositioning, side-effect prediction, poly-pharmacology, and drug resistance [1, 2]. In *in vitro* or *in vivo* experimental approaches, the interaction between drugs and target proteins is measured using the half-maximal inhibitory concentration ($IC_{50}$) or half-maximal effective concentration ($EC_{50}$), inhibition constant ($K_i$), dissociation constant ($K_d$) values. Laboratory experiments that are conducted to measure the affinity value for a large pool of drug-target pairs are time-consuming and expensive. Hence, computational approaches have received even more attention in recent years [3]. In computational-based approaches, feature extraction from the input data is a crucial step in DTI prediction. Feature extraction is a process of determining informative, discriminant, and non-redundant knowledge from raw data such that it facilitates the subsequent learning steps. Feature extraction methods can be divided into two categories: data-driven or non-data-driven (Fig. **1**). The main difference between these two categories is that in data-driven-based approaches, features are learned automatically for each input space. In non-data-driven based approaches, features, for each input, are computed in a fixed way. Deep learning is a class of machine learning methods that learn new hierarchical feature representations with an approach inspired by

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the human visual system [4]. In the last decade, machine learning especially deep learning has significantly impacted many research areas like genomics [5], non-coding RNA classification [6, 7], protein secondary structure prediction [8], De novo drug design [9], bioinformatics [10-12], computer vision [13], natural language processing [14], and language translation [15]. Deep learning is also applied to Drug-target Interaction (DTI) prediction approaches. In this paper, we review and analyze deep learning-based DTI approaches from multiple perspectives.

Pahikkala *et al.* [16] explored the current methods in DTI prediction and found four factors that could impact the prediction results. These factors include 1) problem formulation: DTI prediction can be defined as a classification that assigns an active or inactive label to each drug-target pair or it can be defined as a regression problem to predict the interaction affinity value. Preferably, the DTI prediction is considered as a continuous value prediction problem and after the prediction, to convert it to a classification problem, thresholding is used. In a study [17], for some datasets, predefined thresholds are given. 2) evaluation of the datasets: it determines which dataset is used to evaluate the approach. The properties to choose the dataset are like the diversity of drug compounds or protein targets in the dataset. 3) evaluation procedure: which setting is used to report the final performance measure. For example, simple or nested cross-validation is used to compute the performance and 4) the experimental setting: it indicates the difference between the marginal distribution of training and test datasets. In this case, there are four different settings:

- warm setting: if both compound and protein of unseen pairs in the test set are observed in the training set.
- cold target setting: protein targets of the test set are not seen in the training set.
- cold drug setting: drug compounds of the test set are not observed in the training set and,
- the case where both compounds and protein targets are not observed during the training phase (most challenging setting).

In this paper, in addition to the mentioned factors, we introduce essential and useful factors in designing deep learning-based methods for DTI prediction. To this end, by investigating deep learning-based DTI prediction approaches, the following factors are introduced: 1) the input feature space of the network, 2) feature extraction networks for drug compound and protein target, 3) fusion of drug and protein feature descriptor and 4) prediction layers. In the following, at first, an introduction to the classical neural network is given. Then, each of the mentioned steps is investigated in detail and some guidelines are given. We also introduce drug-target interaction-related datasets and their properties. Next, the important challenges in DTI are given and analyzed and conclusions are given.

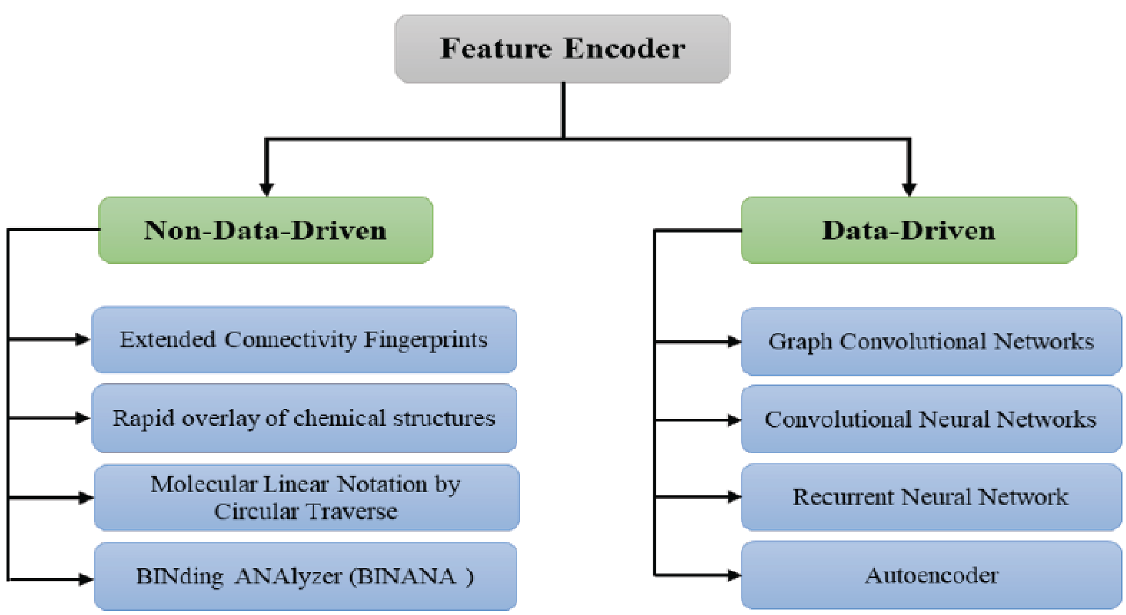


**Fig. (1).** Feature extraction methods are divided into two categories: data-driven or non-data-driven. In each category, four methods are mentioned. *(A higher resolution / colour version of this figure is available in the electronic copy of the article).*

## 2. NEURAL NETWORK

The classical neural networks consist of multiple layers (commonly an input layer, at least one hidden layer, and an output layer) in which each layer contains a set of neurons. Each neuron receives an input vector, then it computes the weighted sum of the input, and finally, it is passed to the activation function to produce the output of the neuron (Fig. **2A**). The activation function can add the non-linearity to the model, while without the activation function, the search space of the network only contains a linear mapping from input to output. Some examples of the activation functions are shown in Fig. **2A**. The weights of each neuron are adjusted during the training phase by optimizing the loss function (Fig. **2B**). The loss function assigns a real number to each output of the network. If the output of the network deviates from the true value, it gets the large number and if the predicted output of the network matches the true value, it gets the lowest number. Classical neural networks can not take into account the local spatial-hierarchies or long-range dependencies of the input [18]. Hence, other networks like CNN and LSTM get more attention to extract the more informative knowledge from the input data (section 4).

## 3. THE INPUT FEATURE SPACE OF THE NETWORK

The input feature space of a deep network can be a raw sequence or an extracted molecular fingerprint or a combination of them. Many works have used a raw molecule sequence as an input of the network. Some works map a raw sequence to another suitable representation like one-hot encoding to feed it into deep networks like CNN or LSTM. In one-hot encoding, each character of the sequence is encoded by a binary vector in which the corresponding bit is set to one, and the other ones are set to zero. From a different perspective, molecular fingerprints are extracted and then are fed as input to the network. In this case, the graph structure of molecules is transformed into a fixed-size feature descriptor. Extended-Connectivity Fingerprints (ECFP) is a topological circular fingerprint in which, for each atom in a compound, the neighboring environment is encoded into a hashed integer identifier that denotes the unique fragment identifier [19]. ECFP is a bit-wise descriptor in which each bit corresponds to a certain substructure in a compound. Ma *et al.* [20] showed that when fingerprints are used as the input to a deep neural network, it gets a significant improvement compared to base approaches like Random Forest.

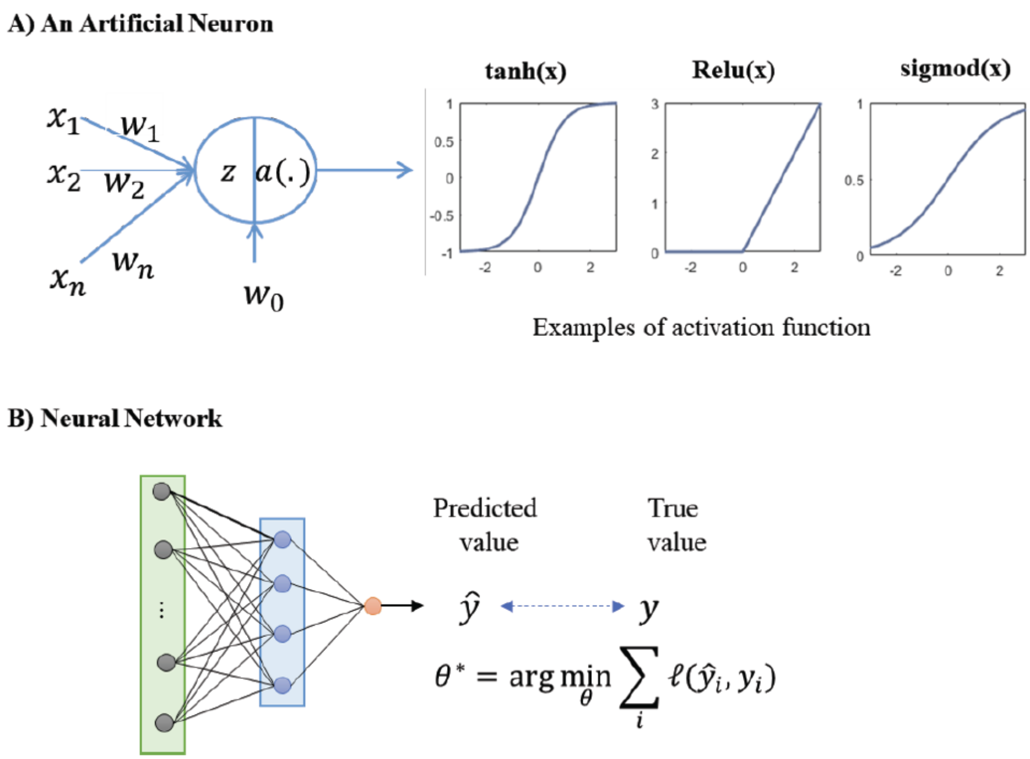


**Fig. (2).** (**A**) a schematic view of an artificial neuron. Also, some examples of activation functions are given. (**B**) a simple neural network with an input layer, one hidden layer, and output layer. The parameters of the model are adjusted by optimizing the loss function. *(A higher resolution / colour version of this figure is available in the electronic copy of the article).*

Chakravarti and Alla [21] proposed a new molecular notation which is called Molecular Linear Notation by Circular Traverse (MLNCT) to feed the deep network. MLNCT travels each heavy atom outwards by one bond in each iteration and records the visiting atoms. Each step of iteration and each atom are respectively separated by a space and an asterisk. In MLNCT, each atom is encoded by some characteristics like atom symbol, hybridization, number of bonded hydrogens, number of double or triple bonds, charge, and ring membership. Durrant & McCammon [22] proposed a novel algorithm for a compound-protein descriptor called BINANA (BINding ANAlyzer). BINANA identifies Close Contacts (within 2.5 A), Electrostatic Interactions (within 4 A), Binding-Pocket Flexibility, Hydrophobic Contacts, Hydrogen Bonds, Salt Bridges, π Interactions and Ligand Atom Types and Rotatable Bonds. There are many other molecular fingerprint approaches, such as coulomb matrix [23], grid featurizer [24], Symmetry function [25], NNscore featurizer [26] and SPLIF [27].

## 4. FEATURE EXTRACTION NETWORKS

By investigating the published literature, we found out that six feature encoding architectures have gotten more attention in DTI prediction which include: 1) Convolutional Neural Network-based (CNN-based), 2) Graph Convolutional Network-based (GCN-based), 3) Recurrent Neural Network-based (RNN-based), 4) Autoencoder based, 5) Deep belief network-based (DBN-based), and 6) Word-embedding based methods. In the following, we introduce each of these architectures and analyze how they are used in DTI prediction.

### 4.1. Convolutional Neural Network

CNN is the most widely used deep neural network which acts on grid-like data structure such as digital images. CNN includes several convolutional and pooling layers that are placed in optional order. Convolutional layers learn a set of filters that extract a set of local patterns in a local receptive field of the layer input. In successive convolutional layers, the receptive field is also enlarged. The pooling layer enlarges the receptive field by downsampling the input of the layer. Also, the pooling layer defines no additional parameter.

The input of the CNN can be 1D or 2D matrices in which the input is scanned only in one or two directions along the sequence, respectively. Up to now, 1D CNNs are applied to DNA sequences for different tasks like DNA sequence classification and DNA-protein binding or motif extraction [28-30]. In (Fig. **3**), the operation of the 1D convolutional layer on protein sequence and a small molecule sequence in SMILE format is shown. In (Fig. **3A**), a protein sequence is shown using a sequence of amino acids that is embedded into a matrix in which each amino acid is encoded by one-hot representation (the corresponding bit of the symbol is one and all of the other bits are zeros). As it is shown, the filter is moved along the sequence and is learned by minimizing the loss function on positive and negative samples of the task. In (Fig. **3B**), the SMILE sequence consists of a sequence of characters. Each character denotes an atom or structure indicator of a molecule. Then, each character is encoded in one-hot representation and is placed in each column of the matrix. In both cases, the learned filters are shown. Tsubaki *et al.* [31] get the protein sequence as input and then feed it into CNN to learn the protein target features. Öztürk *et al.* [32], use the CNN to learn the feature vector for both protein target sequence and drug compound sequence. In [33], the drug compound sequence is described using two methods: In the first method, features are derived by [34] and fed into three fully connected layers and in the second method, the molecule is described using a SMILE matrix and then fed into a CNN. It should be noted that CNN has more explainability and interpretability by visualizing the learned filters and feature maps.

### 4.2. Graph Convolutional Network

In [35, 36], graph convolutional networks for learning molecular fingerprints are introduced. In GCN, the graph structure of a molecule is fed into the network as an input in which each atom is represented by a node and each bond is denoted by an edge. To this end, the graph structure is converted into two matrices: the node descriptor matrix and the adjacency matrix. In the node descriptor matrix, each row shows the corresponding atom descriptor. In (Fig. **4A**), the schematic view of GCN is shown. Commonly, RDKit [37] is utilized to extract the atom's basic features like the atom-type, the number of bonded neighbors in the graph, the number of radical electrons, and implicit valences. Also, the adjacency matrix encodes the neighboring atoms in the molecule. The main layer of a GCN is the graph convolutional layer, which is initially introduced in [38] on spectral representations of graphs. GCN is composed of successive graph convolutional layers that learn effective local substructures of a molecule hierarchically. In this case, for each data, a particular molecular substructure is learned. GCL operates on each node by mapping its feature vector into a new feature space. Then, the feature vector of each node is updated using the feature

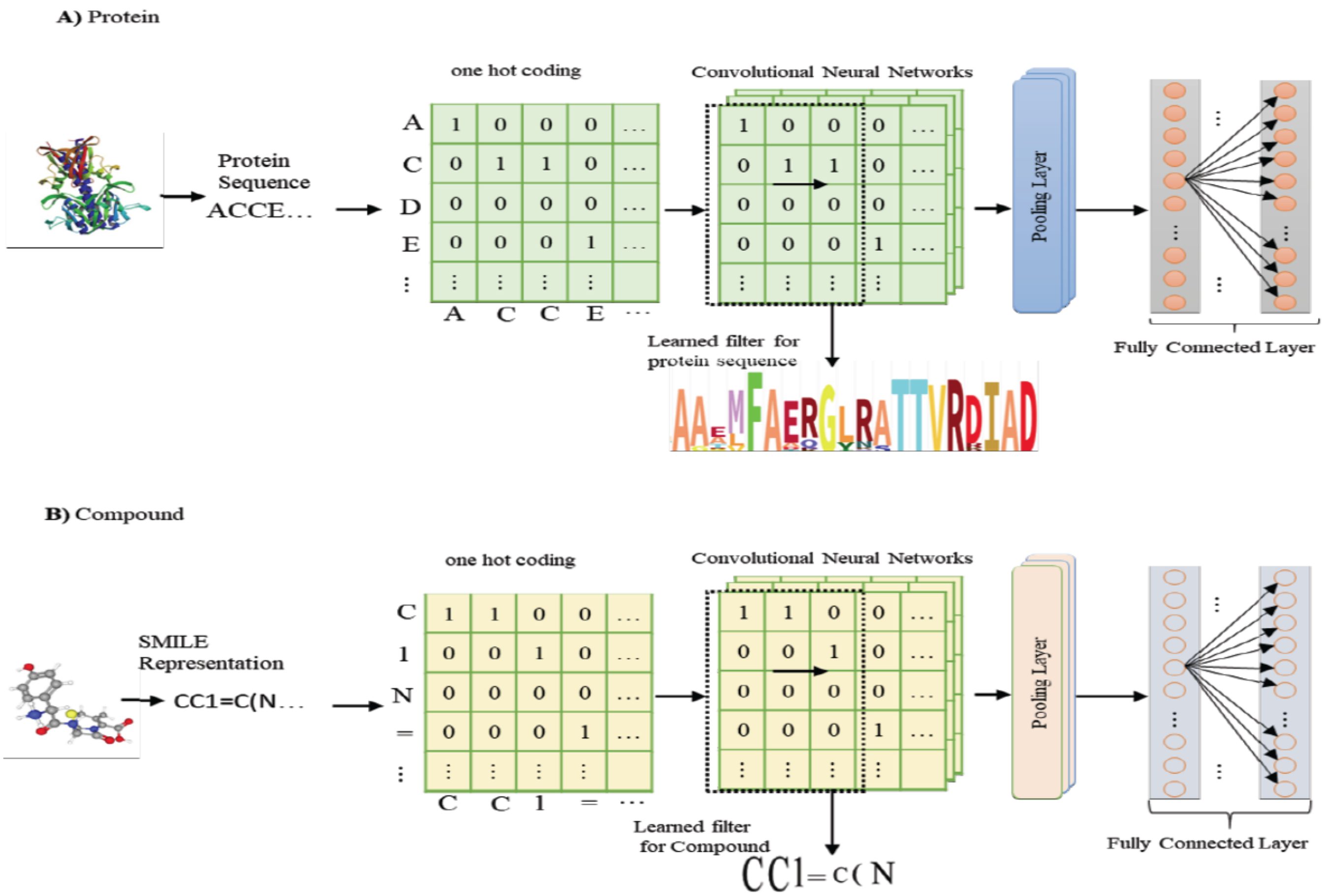


**Fig. (3).** The operation of a 1D convolutional layer on (**A**) a protein sequence and (**B**) a small molecule sequence represented in SMILE format. *(A higher resolution / colour version of this figure is available in the electronic copy of the article).*

vectors of all neighbors of that node. Each successive graph convolutional layer operates on the output of its preceding layer. Therefore, the receptive field on the input molecule is enlarged in the last graph convolutional layers. The graph pooling layer and the graph gather layer are other layers that were initially introduced by Altae-Tran *et al.* [39]. GPL enlarges the receptive filed without defining any parameters. To this end, each node descriptor vector is updated by applying a max operation on each node and its neighbors in the graph structure. The graph gathers layer sums up the feature vector of all nodes into a fixed-size descriptor which describes the whole graph. Pope *et al.* [40] modified a GCN to have more explainability by determining effective functional groups (*i.e.* localized parts of a graph) in the classification task. To determine the functional groups, they utilized four methods: gradient-based saliency maps, class activation mapping (CAM), gradient-weighted CAM (Grad-CAM), and excitation backpropagation. Ryu *et al.* [41] incorporated attention mechanisms and gate mechanisms into GCN. Li *et al.* [42] modified a GCN such that it operates on diverse graph structures which is called Adaptive GCN (AGCN). They propose a new layer in which the graph laplacian is learned by parameterizing the distance metrics. In this case, a unique graph is constructed for input samples of different shapes and sizes. Tsubaki *et al.* [31] used the GCN to learn the drug compound features for DTI prediction. Altae-Tran *et al.* [39] introduced a new framework for low data drug discovery by utilizing a one-shot learning concept. They combined graph convolutional neural networks with the iterative refinement long short-term memory to learn more meaningful features. Gao *et al.* [43] proposed an interpretable approach for DTI prediction in which GCN is utilized to extract learnable features for the drug compound sequence. Wu *et al.* [24] proposed a model called MoleculeNet in which multiple previously proposed molecular featurization and learning algorithms are applied on multiple public datasets and then their results are evaluated and compared. They conclude that graph-based methods (like GCN) significantly outperform the other comparable methods which shows that they learn more meaningful features. Compared to

CNN, the interpretability of GCN is low. Although some works to improve the interpretability power of GCN are being done. Pope *et al.* [44] developed three methods to improve the interpretability of GCN including the contrastive gradient-based (CG) saliency maps, Class Activation Mapping (CAM), and Excitation Backpropagation (EB) and their variants, gradient-weighted CAM (Grad-CAM) and contrastive EB (c-EB). GCN has more advantage compared to CNN, since GCN utilize the graph structure of the molecule, however, CNN utilize a 1D sequence.

### 4.3. Recurrent Neural network

RNN is a class of neural networks that takes a time-series as input. During the training, it remembers the past to utilize it in decision making for the next step. RNN is composed of a series of recurrent cells where each cell includes hidden gates that provide state information to pass to the next cell. RNN in large sequential data faces the forgetting problem and gradient vanishing. To cope with these challenges, Long Short Term Memory (LSTM) is introduced. Each cell of LSTM includes an input gate, a forget gate, and an output gate that learns what to forget and what new information should be preserved to pass to the next cell [45]. The schematic operation of LSTM on a molecule is represented in Fig. (**4B**). In [21], MLNCT descriptors are fed into a bidirectional LSTM as the input. Fooshee *et al.* [46] fed the SMILE representation of the molecule into LSTM to predict electron sources and sinks.

### 4.4. Autoencoders

Autoencoders are a class of neural networks that learn the feature encoder of the input domain in an unsupervised manner [47]. It is composed of two subnetworks: feature encoder and decoder (Fig. **4C**). Autoencoder learns to compress the input into low-embedded space through an encoder subnetwork such that it can be reconstructed back into the input from the low-embedded space through a decoder subnetwork. Karimi *et al.* [48] used seq-2-seq autoencoders to get the representation space for compounds and proteins. In seq2seq autoencoders, recurrent units are used in the encoder and the decoder subnetworks [49]. Xu *et al.* [50] proposed an autoencoder based architecture to embed the molecule SMILE into a vector. They used a multilayered Gated Recurrent Unit (GRU) network as an encoder and decoder.

### 4.5. Word Embedding

Word embedding is a feature learning technique that was initially introduced in natural language processing (NLP). It learns to map words or phrases of the dictionary into the real-valued vector such that the context of a word in a document, the semantic and the syntactic similarity and its relation with other words are preserved. In recent years, extending word embedding techniques have gotten considerable attention in NLP approaches like frequency-based embedding (count vector, TF-IDF, and co-occurrence vector), prediction-based embedding (word2vec [51], GLoVe [52], ULMFiT [53]). Asgari *et al.* [54, 55] have utilized the word embedding techniques to provide a new representation for biological sequences. Öztürk *et al.* [56] extract a set of chemical words from SMILES representations of the compounds. Then, a protein is represented using a set of chemical words in their strong binding ligands using the highest Term Frequency- Inverse Document Frequency (TF-IDF) technique. Öztürk *et al.* [32] utilized the word embedding mechanism to embed the protein or compound sequence to feed into the subsequent layers.

### 4.6. Deep Belief Networks (DBN)

Deep Belief Networks (DBN) is a class of neural networks that are created by stacking restricted Boltzmann machines (RBMs), which learn to extract a deep hierarchical feature from the data. RBM is a bipartite graphical model that consists of two layers: one visible layer and one hidden layer. Each node in the visible layer is connected to each node in the hidden layer. In RBM, the joint probability distributions over binary hidden and visible variables are defined in terms of an energy function, and the corresponding parameters are obtained by minimizing it. In unsupervised learning of RBM, instead of reducing the reconstruction error, the joint probability is maximized. To learn parameters in DBN, first, each RBM is learned layer-wise in an unsupervised manner and then the whole network is fine-tuned through supervised learning. Wen *et al.* [57], used an Extended Connectivity Fingerprint (ECFP) to describe drug compound and Protein Sequence Composition descriptors (PSC) to describe the protein target. Then, these vectors are concatenated and fed into the DBN.

## 5. THE FUSION OF DRUG AND PROTEIN FEATURE DESCRIPTOR

In this section, the strategies to fuse drug and protein feature descriptors are investigated. Commonly, these descriptors are concatenated to form the final descriptor. In [33], these two descriptors are concatenated to predict the compound property. However, in recent years, the attention-based mechanism is also utilized to better fuse drug and protein feature descriptors. The attention-based mechanism was initially introduced by Bahdanau *et al.* [58]. Tsubaki *et al.* [31] used the attention mechanism to compute the interaction strength between a molecule and a set of protein subsequences. Then, the weighted sum of protein subsequence's features with attention coefficients is concatenated with a compound descriptor to be fed into the prediction layers.

Hassan *et al.* [59] utilized a BINANA to describe compound-protein interaction and then fed it into a deep neural network to predict the interaction affinity. Note that contrary to the other methods, in [59], the input feature of the network is a combination of compound and protein interaction features. In Table **1**, the summary of some approaches in deep learning-based DTI prediction is given. Abbasi *et al.* [60] introduced an approach in DTI prediction, which is called DeepCDA. They utilize CNN-LSTM as a feature descriptor. Their main contribution is to utilize the domain adaptation technique to learn a feature descriptor for unseen drug-target pairs.

## 6. DATASETS

In this section, we review the existing datasets in DTI and then we analyze the characteristics of each. DTI datasets are stores of information about compound sequences, protein sequences, and binding affinity values. The binding affinity value is usually expressed in different measures like $IC_{50}$, $EC_{50}$, $K_i$, and $K_d$. These various sources may cause some challenges. For example, if a model is trained on a dataset with $IC_{50}$ binding affinity and is tested on a different dataset with different binding affinity values, then evaluating that model is going to be a challenge. Hence, Tang *et al.* [64] introduced a new score called KIBA score that combines affinity measures from different sources like $IC_{50}$, $K_i$, and $K_d$ into a new measure called KIBA measure. In Table **2**, A summary of the investigated datasets in this paper is given. In the following, each of these datasets discussed in detail and compared to the others.

Antifungal Synergistic Drug Combination Database (ASDCD) is a dataset which facilitates the synergistic analysis of antifungal drug combination [65]. STITCH 5 is a large dataset that contains 1.6 billion interactions for over 340000 chemicals and over 9.6 million proteins from 2031 eukaryotic and prokaryotic genomes, and it provides the interaction vector for each drug [66]. In STITCH 5, a new feature is added, which allows a user to filter the network to show the results on proteins in a specified tissue [66]. This property can help the researchers to utilize similar drug-target interactions for unseen pairs.

BindingDB is an online database that contains over one million interaction data records, derived from various resources. It provides links to PDB, PubMed, pathway information, and other resources [67].

The KIBA dataset investigates the effect of a given perturbagen on kinase inhibitor bioactivities [64]. It originally contains 246,088 drug-target interaction with 52,498 drugs and 467 targets, in which all drugs and targets with less than 10 interactions are removed [17]. The Davis dataset contains over 30,000 drug-target interactions for 68 unique drugs and 442 unique proteins which affinity measured by the $K_d$ value. Similar to the KIBA dataset it also reports affinity values for the kinase protein family [17, 68]. The Metz dataset includes 1421 drugs and 156 targets in which the binding affinity is given as the $pK_i$ value [69].

Toxicity Forecaster of EPA generates data and predictive models on thousands of chemicals of interest and gathers them into the Toxcast dataset. This dataset contains qualitative results of over 600 experiments on 8615 compounds [70].

DrugBank was first released in 2006 and has gradually evolved over recent years. The summary of the coverage of different versions of DrugBank datasets is given in [71]. DrugBank covers four types of protein target families including GPCRs, enzymes, nuclear receptors, and ion channels as well as SuperTarget and BindingDB datasets.

As it is shown in Table **2**, the diversity of drugs and targets in various datasets is different. BindingDB has more diversity of drugs than all the other datasets and in the protein targets, STITCH has the most diversity. Moreover, each of these datasets covers only some protein target family and drug compounds. By investigating deep learning-based approaches, we found that some of these datasets like KIBA, Davis, Metz, BindingDB, and PDBind, have more applicability in this research area. Some of these datasets like BindingDB, STITCH, and DrugBank are frequently updated, while the other datasets remain the same for several years. Due to a large number of available datasets, it is necessary that future researches provide more information about the intersection between them and help the researchers to select the most appropriate datasets for their research. Moreover, the different numbers of available interactions for some drugs or targets lead to data heterogeneity challenges.

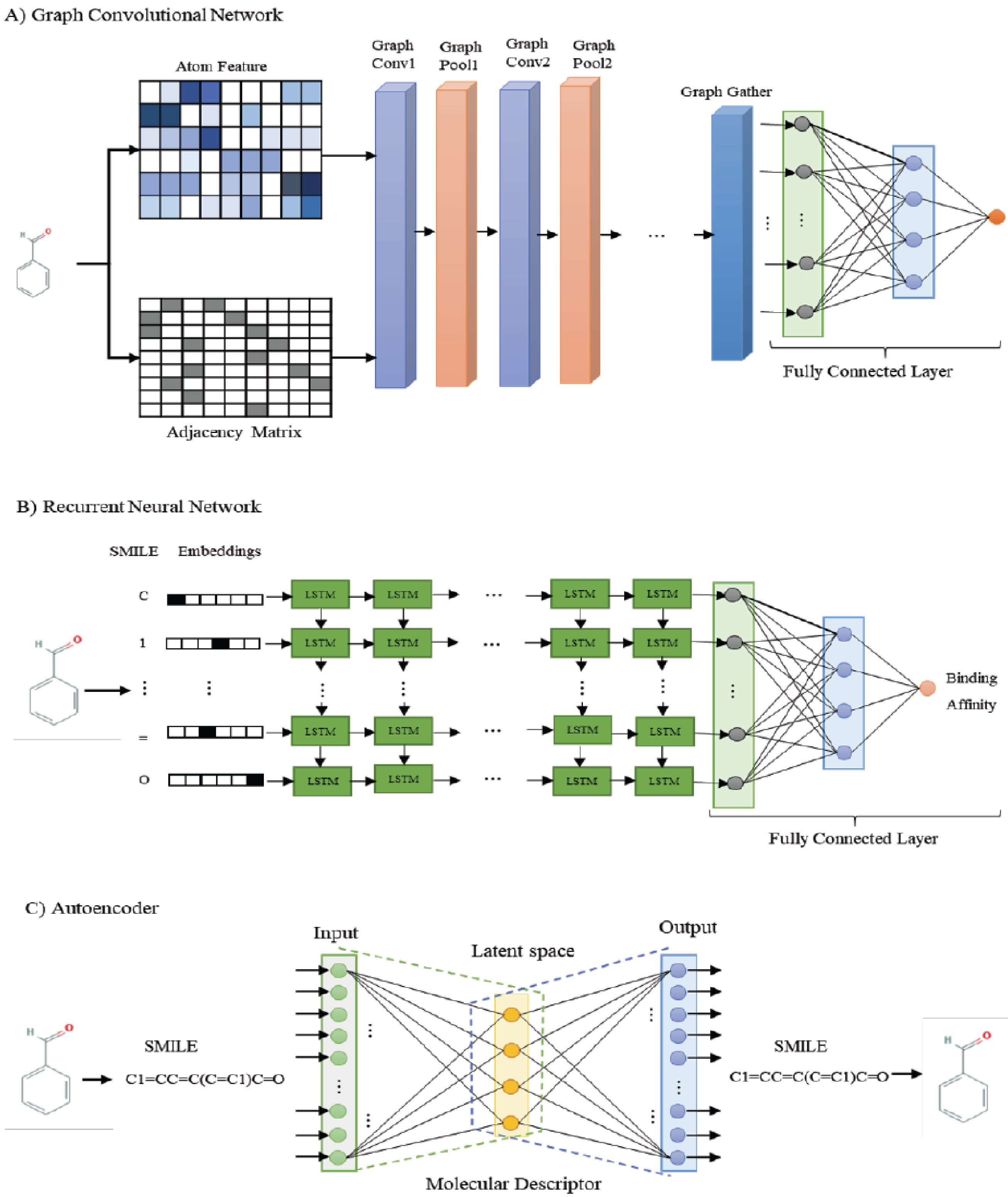


**Fig. (4).** Architectures of deep learning models **A**) Graph Convolutional Network: a matrix of atom descriptors and adjacency matrix (bonds between atoms) are fed as the input to the network. **B**) Recurrent Neural Network: each character of SMILE is fed into one cell of the LSTM. Here, the stacked layers of the LSTM are shown. **C**) Autoencoders: the encoder gets a molecule sequence as an input and maps it to a low dimensional space and the decoder maps this new representation into the input space. *(A higher resolution / colour version of this figure is available in the electronic copy of the article).*

**Table 1. A summary of the most common approaches in deep learning-based DTI prediction.**

| Approach | Year | Protein Descriptor | Compound Descriptor | Fusion | Prediction network | Datasets |
|---|---|---|---|---|---|---|
| DeepCDA[60] | 2020 | CNN-LSTM | CNN-LSTM | Simple concatenation | ADAN* | Davis, KIBA, BindingDB |
| DeepDTA [32] | 2019 | CNN | CNN | Simple concatenation | MLP | Davis, KIBA |
| DeepAffinity [48] | 2018 | Unied RNN-CNN | Unied RNN-CNN | Simple concatenation | MLP | BindingDB |
| CPI prediction with end-to-end learning [31] | 2018 | CNN | GCN | One-sided Attention Mechanism | MLP | DTI datasets, for human and C.elegans [61] |
| PADME [62] | | PSC descriptor | GCN | Simple concatenation | MLP | Davis, KIBA, Metz, Toxcast |
| DLSCORE [59] | 2018 | BINANA algorithm | BINANA algorithm | The input features are computed for the interaction sites of drug-target pair | DNN | PDBind |
| MT-DTI [63] | 2019 | CNN | Self-Attention+ molecule token embedding (MTE) and positional embedding (PE) | Simple concatenation | MLP | Davis, KIBA |

*Adversarial Domain Adaptation Network

**Table 2. The summary of the datasets in DTI.**

| Dataset | No. Drug | No. Target | No. Drug-Target Interactions |
|---|---|---|---|
| ASDCD | 105 | 1225 | 210 |
| BindingDB | 780,240 | 7,371 | 1,756,093 |
| DrugBank | 11,682 | 26,889 | 131,724 |
| SIDER | 5868 | 1430 | 139,756 |
| STITCH | 430,000 | 9,600,000 | 1,600,000,000 |
| SuperPred | 341,000 | 1,800 | 665,000 |
| SuperTarget | 195,770 | 6219 | 332,828 |
| Davis | 442 | 68 | 30056 |
| KIBA | 229 | 2111 | 118254 |
| Metz | 1423 | 170 | 35259 |
| ToxCast | 8575 | 617 | 530605 |

## 7. DISCUSSION AND FUTURE WORK

In this section, we explore the challenges in DTI prediction and discuss the recent advances in deep learning proposed to overcome them. As mentioned earlier, a crucial step in DTI prediction is the feature extraction step. Up to now, we have seen several studies. However, further experimental studies are necessary to compare different feature extraction methods. Moreover, it is shown that the attention mechanism has an important effect on the final performance, due to the two-sided effect of input substructures that are being evaluated. From a biological perspective, local substructures or fragments have an important role in protein's function. Wang *et al.* [72] proposed an approach called DeepFragLib, which builds a high-quality fragment library using deep contextual learning techniques. It is hoped that by incorporating deep-based fragment extraction methods into DTI prediction approaches, the efficacy of finding important features can be raised.

One of the other key challenges is to increase the generalizability of the DTI prediction model. A model with higher generalizability provides more reliable results on unseen test data. As mentioned earlier, the experimental setting in DTI prediction can be divided into four categories. By investigating the published methods, it is determined that a warm setting gets the most attention. Among the other categories, the last category, in which the training dataset and the test dataset have completely different distributions, has attracted the least attention. Deep transfer learning tries to sketch out how to transfer knowledge from a learned model to another model [73-75]. In other words, it learns a model using a similar auxiliary data (source task) and then utilizes them to learn a new model for a new task (target task). Also, it is assumed that source and target tasks might have different distributions (marginal or conditional), while their domains are the same. In deep transfer learning, domain adaptation techniques are utilized to map source and target data into a new representation such that they have the same distribution. Up to now, deep transfer learning in DTI prediction gets the least attention. Abbasi *et al.* [76] proposed a model to learn a compound representation for low data drug discovery, which is transferable across domains and tasks. They assumed that source and target assays are adapted to different protein target groups with different compounds distribution. They utilized a multi-task network for training the source model. They used a GCN as a feature extraction network. Then, the target encoder network is learned using an improved version of the adversarial domain adaptation technique [77]. Moreover, they proposed a new term called semantic transfer to consider the discriminative ability of unlabeled target samples. To our best knowledge, this is the only study that is done to overcome this challenge. Hence, more studies should be done to develop stronger and more reliable methods to cope with this challenge.

Most of deep learning-based DTI predictions are based on structural similarities of drug-target pairs rather than their biological relevance. However, Ball *et al.* [78] showed that, in the read-across methodology, incorporating the similarity between drugs and between targets is essential to predict the interaction of unseen drug-target pairs. Also, Chen *et al.* [79] proposed a model in which for the first time, Gene Ontology (GO) and KEGG enrichment scores were incorporated in DTIs prediction. Their results showed that this knowledge is important in drug target-based class identification. Gao *et al.* [43] proposed a deep learning-based DTI approach. In their approach, not only protein and compound sequences are incorporated but also a set of GO annotations for each protein is utilized. It should be noted that GO annotation provides high-level information about the function of the protein. Their obtained results showed that GO annotation has a significant impact on the final performance. However, to our best knowledge, none of the deep learning-based methods has been applied to jointly consider this knowledge in DTI prediction.

## CONCLUSION

In this study, we have provided an overview of deep learning-based DTI prediction approaches. To this end, we explored methods from three viewpoints: input feature space, feature extraction methods, and fusion methods. In the feature extraction step, we investigate five common deep architectures that are used to describe protein and compound sequences. Also, the advantages and disadvantages of each architecture are discussed. Moreover, we discussed how the existing approaches combine drug compound and protein target feature descriptors. It is figured out that attention-based fusion approaches provide more discriminative knowledge than the simple concatenation approaches. Based on this overview, we found some main challenges: incorporating deep-based fragment extraction by utilizing the transfer learning concept, incorporating extra knowledge such as the similarity between drugs and between targets, GO and KEGG enrichment, and incorporating deep-based fragment extraction into DTI approaches. In this work, we provide some guidance to conduct future research in deep learning-based DTI prediction.

## LIST OF ABBREVIATIONS

| | | |
|---|---|---|
| AGCN | = | Adaptive GCN |
| ASDCD | = | Antifungal Synergistic Drug Combination Database |
| BINANA | = | BINding ANAlyzer |
| CAM | = | Class activation mapping |
| CNN | = | Convolutional Neural Network |
| DBN | = | Deep belief network-based |
| DTIs | = | Drug-target interactions |
| $EC_{50}$ | = | Half-maximal effective concentration |

ECFP = Extended connectivity fingerprint
GCN = Graph Convolutional Network-based
GO = Gene Ontology
Grad-CAM = Gradient-weighted CAM
$IC_{50}$ = Half-maximal inhibitory concentration
$K_d$ = Dissociation constant
$K_i$ = Inhibition constant
LSTM = Long Short Term Memory
NLP = Natural language processing
PSC = Protein sequence composition
RBM = Restricted Boltzmann machines
RNN = Recurrent Neural Network
TF-IDF = Term Frequency- Inverse Document Frequency

## CONSENT FOR PUBLICATION

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## CONFLICT OF INTEREST

The authors declare no conflict of interest, financial or otherwise.


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